

Chapter 26: Induction and Augmentation of Labor —

Revision Notes

Williams Obstetrics 26e, pp. 1244–1272 of the PDF. High-yield numbers are in **bold**. The six tables are transcribed from the PDF images (they are missing from the extracted chapter text). Both figures are included from the PDF.

Big picture:

- **Induction** = stimulating contractions **before spontaneous labor**, with or without ruptured membranes.
- **Cervical ripening** = softening and opening a closed, uneffaced cervix before induction (it can itself induce labor).
- **Augmentation** = enhancing spontaneous contractions that are inadequate (failed dilation, poor descent).
- Know: **ARRIVE**, the **Bishop score**, **dinoprostone vs misoprostol vs Foley**, **oxytocin regimens**, and amniotomy.

1. Labor Induction

Indications and contraindications

- US induction rate: **9.5% (1991) → 27% (2019)**.
- Induce when the **benefits to mother or fetus outweigh continuing the pregnancy**.
- **Common indications**: ROM without labor, **gestational hypertension**, **oligohydramnios**, **nonreassuring fetal status**, **postterm pregnancy**, maternal disease (**chronic hypertension**, **diabetes**).
- **Contraindications** = anything that precludes spontaneous labor or vaginal delivery:

Maternal	Fetal
Abnormally implanted placenta (previa, accreta)	Appreciable macrosomia
Prior uterine incision with high rupture risk (e.g., classical)	Severe hydrocephalus
Active genital herpes	Malpresentation
Contracted or distorted pelvis	Nonreassuring fetal status
Cervical cancer	

Risks

- **Chorioamnionitis**, **uterine rupture**, **PPH from atony**.
- **Cesarean**: older nonrandomized studies suggested ↑ risk in induced nulliparas; newer ones show **comparable or lower** rates vs spontaneous labor. **Expectant management is the better comparator** than spontaneous labor.
- **ARRIVE trial** (>6000 low-risk nulliparas at 39 wk, induction vs expectant):
 - Primary outcome (**severe neonatal morbidity or death**): **no difference**.
 - **Cesarean 18.6% vs 22.2%** (significantly lower with induction).
 - Chorioamnionitis, PPH, peripartum infection: **no difference**.
 - Meta-analysis of 6 cohort studies: 39-wk elective induction → **lower cesarean** and morbidity.

Table 26-1. Selected Maternal and Perinatal Outcomes with Labor Induction Versus Expectant Management in Low-Risk Nulliparous Women — the ARRIVE Trial

Factor	Induction (%) (n = 3059)	Expectant (%) (n = 3037)	p value
Maternal outcome^a			
Cesarean delivery	18.6	22.2	<0.001
Operative vaginal delivery	7.3	8.5	0.07
Hypertensive disorder	9.1	14.1	<0.001
Chorioamnionitis	13.3	14.1	0.35
Postpartum hemorrhage	4.6	4.5	0.81
Perinatal outcome^b	4.3	5.4	0.049
Respiratory support	3.0	4.2	
Apgar score ≤ 3 at 5 min	0.4	0.6	
Seizure	0.4	0.1	
Meconium aspiration	0.6	0.9	
Birth trauma	0.5	0.6	
Perinatal death	0.1	0.1	

^aSecondary outcomes. ^bComposite outcomes. ARRIVE = A Randomized Trial of Induction Versus Expectant Management. The composite perinatal outcome (the trial's primary outcome) shows $p = 0.049$ in the table, while the text says it "did not differ" between groups.

Uterine rupture and other risks

- Rupture risk is low overall but **highest with a prior uterine scar**.
- ⚠️ **ACOG: no prostaglandins** for ripening or induction **with a prior uterine incision**. **Oxytocin induction is an option** but carries a higher rupture rate and lower VBAC rate. Oxytocin **augmentation** may be used in TOLAC (evidence on rupture conflicting).
- **Amniotomy** → **more chorioamnionitis** than spontaneous labor.
- Atony/PPH more common with induction or augmentation in some studies. Parkland: induction linked to **17% of 553 unanticipated peripartum hysterectomies**.

Elective induction

- ACOG previously advised **against elective term induction for convenience** (exceptions: risk of rapid labor, psychosocial reasons, distance from hospital/family support).
- ⚠️ **Elective delivery before 39 completed weeks** → **significant neonatal morbidity**.
- **After ARRIVE, ACOG (2018) and SMFM (2019): offering elective induction to low-risk, well-dated nulliparas at 39 wk is reasonable** — they must meet ARRIVE criteria, with **shared decision-making** and consideration of **resources**.

Factors affecting induction success

- **Favorable: younger age, multiparity, BMI <30, favorable cervix, birthweight <3500 g**.
- **Consortium on Safe Labor: elective induction** → vaginal delivery in **97% of multiparas vs 76% of nulliparas**; better with a ripe cervix.
- Failed induction depends strongly on **how long you allow**, especially with an unfavorable cervix. **Higher BMI** and **diabetes** lengthen time to active phase and full dilation.

Author	Recommendation before calling induction "failed"
Simon and Grobman (2005)	A latent phase of up to 18 h lets most induced women deliver vaginally without ↑ morbidity
Rouse (2000)	At least 12 h of oxytocin after membrane rupture
Kawakita (2016)	Up to 15 h for multiparas

2. Preinduction Cervical Ripening

- A soft, **"ripe/favorable"** cervix predicts success, though favorability is often subjective.
- Pharmacological and mechanical methods can ripen the cervix and may start labor on their own. But **few data show they lower cesarean rates or morbidity.**

Table 26-2. Some Commonly Used Regimens Compared with Oxytocin Infusion for Preinduction Cervical Ripening and/or Labor Induction

Techniques	Agent	Route/Dose	Compared with Oxytocin
Pharmacological			
Prostaglandin E ₂	Dinoprostone gel, 0.5 mg (Prepidil)	Cervical 0.5 mg; repeat in 6 h; permit 3 doses total	1. Shorter I-D times with oxytocin infusion than oxytocin alone
	Dinoprostone insert, 10 mg (Cervidil)	Posterior fornix, 10 mg	1. Insert has shorter I-D times than gel. 2. 6–12 h interval from last insert to oxytocin infusion
Prostaglandin E ₁ ^a	Misoprostol tablet, 100 or 200 µg (Cytotec) ^b	Vaginal, 25 µg; repeat 3–6 h prn. Oral, 50–100 µg; repeat 3–6 h prn	1. Contractions within 30–60 min. 2. Success comparable to oxytocin for ruptured membranes at term and/or favorable cervix. 3. Tachysystole common with vaginal doses >25 µg
Mechanical			
Transcervical 36F Foley catheter	30-mL balloon	Transcervical	1. Improves Bishop scores rapidly. 2. 80-mL balloon more effective. 3. Combined with oxytocin infusion is superior to PGE ₁ vaginally. 4. With EASI, results improved and possible decreased infection rate
Hygroscopic dilators	Laminaria, hydrogel	Intracervical	1. Rapidly improves Bishop score. 2. May not shorten I-D times with oxytocin. 3. Uncomfortable, requires speculum and placement on an examination table

^aOff-label use. ^bTablets must be divided for 25- and 50-µg dose, but drug is evenly dispersed. EASI = extraamniotic saline infusion at 30–40 mL/h; I-D = induction-to-delivery. Two mismatches with the text: the table lists a "36F" Foley, while the Fig 26-2 legend describes a 26F catheter; and the table gives a 6–12 h interval from the insert to oxytocin, while the text (manufacturer) says 6–12 h after the gel but only ≥30 min after removing the insert.

Cervical favorability — Bishop score

- Higher score → higher rate of vaginal delivery after induction.
- **Bishop score >8 → high likelihood of successful induction. ≤6 → unfavorable** (ACOG).

Table 26-3. Bishop Scoring System Used for Assessment of Inducibility

Score	Dilatation (cm)	Effacement (%)	Station (–3 to +2)	Consistency	Position
0	Closed	0–30	–3	Firm	Posterior
1	1–2	40–50	–2	Medium	Midposition
2	3–4	60–70	–1	Soft	Anterior
3	≥5	≥80	+1, +2	—	—

Memory hook: "Call PEDS" = **C**onsistency, **P**osition, **E**ffacement, **D**ilation, **S**tation. Only dilation, effacement and station can score 3 (max total **13**).

- **Simplified Bishop score** (Laughon, 5610 nulliparas at 38–42 wk): **dilation, station and effacement** alone predict as well or better; consistency and position can be dropped.
- **Transvaginal cervical length** as an alternative: meta-analysis of **31 trials** → **low sensitivity and specificity**, limited value.

3. Pharmacological Ripening Agents

- All are **prostaglandin analogues**; they can also stimulate contractions. Oxytocin is usually started with or after the agent.

Prostaglandin E₂ (dinoprostone)

Form	Dose / use	Details
Prepidil gel	0.5 mg intracervical (2.5-mL syringe), just below the internal os	Supine, stay reclined 30 min . Repeat every 6 h , max 3 doses in 24 h
Cervidil insert	10 mg , releases 0.3 mg/h (slower than gel)	Single dose, placed transversely in the posterior fornix ; little or no lubricant (coats the device). Recumbent ≥2 h . Remove after 12 h , at labor onset, and ≥30 min before oxytocin . Removable by its tail if tachysystole or FHR changes
20-mg suppository	⚠ NOT for cervical ripening	For termination at 12–20 wk and evacuation after fetal demise up to 28 wk

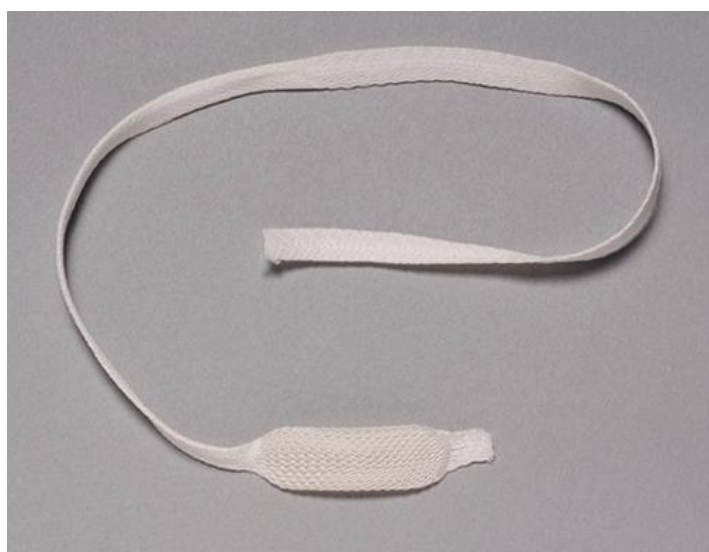


Figure 26-1. Cervidil insert: 10 mg dinoprostone releasing ~0.3 mg/h over ~10 h; the mesh sac's long tail allows quick removal.

Efficacy

- ↑ **vaginal delivery within 24 h**, but **cesarean rate not consistently reduced**.
- **Cochrane (70 trials, 11,487 women)**: more vaginal births in 24 h; **3× ↑ tachysystole with FHR changes**; cesarean not reduced.
- Intracervical gel: ↓ cesarean only in women with an **unfavorable cervix and intact membranes**.
- **PROBAAT-P and -M (Foley vs vaginal PGE₂)**: no difference in cesarean.

Side effects

- **Uterine tachysystole in 1–5%** after vaginal PGE₂.
- **Tachysystole (ACOG) = >5 contractions in 10 min**, always qualified by the **presence or absence of FHR abnormalities**. ⚠ "Hypertonus", "hyperstimulation" and "hypercontractility" are **no longer defined — don't use them**.
- **Don't use prostaglandins in women already in spontaneous labor** (tachysystole with fetal compromise).
- Tachysystole after the insert → **pull it out. Irrigating out gel doesn't help**.
- Manufacturer cautions: **ruptured membranes, glaucoma, asthma** (but in 189 asthmatics dinoprostone did **not** worsen asthma).
- **Manufacturer contraindications**:
 - Dinoprostone hypersensitivity
 - Suspected fetal compromise or **CPD**
 - **Unexplained vaginal bleeding**
 - **≥6 prior term pregnancies**
 - Contraindication to vaginal delivery
 - Already on oxytocin, or oxytocin contraindicated
 - Endangered by prolonged contractions — **prior cesarean or uterine surgery**

Administration

- Give **only in or near the delivery suite**, with **uterine activity and FHR monitoring**.
- Contractions usually appear in the **1st hour**, peak in the **first 4 h**.
- **Oxytocin afterwards: wait 6–12 h after gel, or ≥30 min after insert removal**.

Prostaglandin E₁ (misoprostol, Cytotec)

- Approved as **100- or 200-µg tablets for peptic ulcer prevention**; split for **25- or 50-µg** doses (drug evenly distributed). Vaginal, oral and buccal absorption.
- Widely and safely used **off-label**; stable at room temperature; **preferred ripening prostaglandin at Parkland**.

Vaginal misoprostol

- **Equivalent or superior** to intracervical/vaginal PGE₂.
- vs oxytocin or dinoprostone: **↑ vaginal delivery within 24 h**; ↑ tachysystole but **no change in cesarean**.
- vs dinoprostone: **less need for oxytocin**, but **more meconium-stained fluid**.
- Higher doses → less oxytocin but **more tachysystole** (± FHR changes).
- **ACOG: 25 µg vaginally** (quartered 100-µg tablet).

Oral misoprostol

- Faster peak and decline than vaginal.
- **Cochrane (76 trials)**: vs placebo → ↑ vaginal birth in 24 h, ↓ oxytocin, ↓ **cesarean**. Also ↓ cesarean vs **oxytocin** and vs **dinoprostone**.
- Oral ≈ vaginal in efficacy, but oral → **higher Apgar scores and less PPH**.

Table 26-4. Randomized Trials with 200 or More Subjects Designed to Compare Misoprostol (PGE₁) With or Without Foley Bulb

Study	Comparators	Primary Outcome	N	Findings
Kehl (2015)	Double balloon catheter + oral PGE ₁ (50 µg) vs oral PGE ₁ alone	TTD	326	Longer TTD with Foley + oral PGE ₁ (32 h) vs oral PGE ₁ alone (22 h) (p = 0.004)
Ten Eikelder (2016)	PROBAAT-II: Oral PGE ₁ (50 µg) vs Foley	Neonatal asphyxia or EBL >1000 mL	1859	No difference in primary outcome: Foley (12%) vs oral PGE ₁ (11.5%)
Levine (2016)	FOR MOMI: Vaginal PGE ₁ (25 µg) + Foley vs oxytocin + Foley vs vaginal PGE ₁ alone vs Foley alone	TTD	492	Shorter TTD with combinations: Foley + PGE ₁ (13.1 h), oxytocin + Foley (14.5 h), PGE ₁ alone (17.6 h), Foley alone (17.7 h) (p <0.001)
Kruit (2016)	Foley vs oral PGE ₁ (50–100 µg)	CD and infections	202	No difference in CD or maternal/neonatal infection rates
Mundle (2017)	Foley vs oral PGE ₁ (25 µg)	VD within 24 h	602	VD within 24 h more common with oral PGE₁ (57%) vs Foley (47%) (p = 0.014)
Al-Ibraheemi (2018)	Vaginal PGE ₁ (25 µg) ± Foley	TTD	200	TTD in combination group (15 h) vs vaginal PGE ₁ alone (19 h) (p = 0.001)
Young (2020)	Oral PGE ₁ (50 µg) vs vaginal PGE ₁ (25–50 µg) vs dinoprostone gel (1 to 2 mg)	TTD	511	TTD differences not significant: oral PGE ₁ (22.6 h), vaginal PGE ₁ (25.5 h), vaginal dinoprostone (20.1 h) (p = 0.46)
Adhikari (2020)	Oral PGE ₁ (100 µg) ± Foley	VD	2227	No difference in VD rates: Foley + oral PGE ₁ (78%), PGE ₁ alone (77%)

CD = cesarean delivery; EBL = estimated blood loss; N = participant number; PGE₁ = prostaglandin E₁; TTD = time to delivery; VD = vaginal delivery. The Kehl row prints "oral E₁ alone"; PGE₁ is meant.

Nitric oxide donors

- NO likely mediates cervical ripening (cervical NO metabolites ↑ at contraction onset).
- **Isosorbide mononitrate, glyceryl trinitrate: less effective than prostaglandins, no reduction in cesarean, and more headache, nausea and vomiting.**

4. Mechanical Ripening Techniques

- Options: **transcervical Foley (± EASI), hygroscopic dilators, membrane stripping.**
- **Meta-analyses:**
 - vs prostaglandins → **less tachysystole, same cesarean rate**
 - vs oxytocin → **lower cesarean rate**
 - vs dinoprostone → **more multiparas undelivered at 24 h**
 - Foley vs dinoprostone insert → same cesarean rate, less tachysystole

Transcervical catheter

- Used with an **unfavorable cervix** (it falls out as the cervix opens). OK with **intact or ruptured** membranes.
- Foley through the **internal os**, traction by taping to the thigh.
- **EASI:** constant saline infusion into the space between the internal os and membranes. **Chorioamnionitis 6% vs 16%** without infusion (Karjane). Meta-analysis: **transcervical catheters don't raise maternal or fetal infection.**

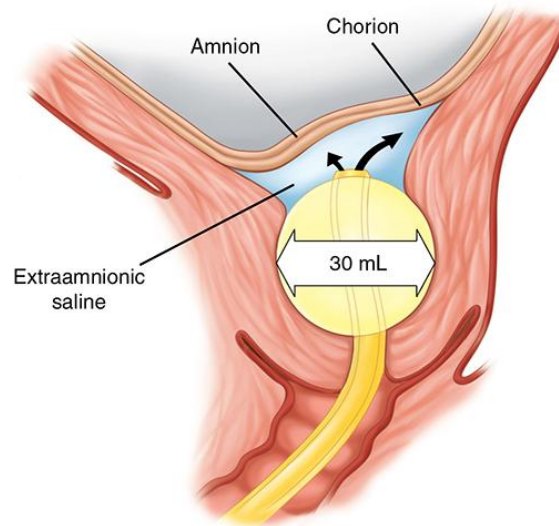


Figure 26-2. EASI: 26F Foley, 30-mL balloon pulled snug against the internal os, room-temperature saline at 30–40 mL/h into the extraamniotic space.

Trial evidence

- **PROBAAT I, P, M, II** (Foley vs dinoprostone gel, insert, vaginal/oral misoprostol): **similar outcomes; fewer CTG changes** with Foley.
- **Foley + concurrent oxytocin** (Schoen): **shorter median time to delivery** than Foley then oxytocin; **cesarean unchanged**. Same with intact membranes (Connolly).
- **Ruptured membranes**: Foley + oxytocin **no better than oxytocin alone** (Amorosa).
- **Foley + misoprostol**: shorter time to delivery, cesarean unchanged. With a catheter, **vaginal misoprostol beat buccal**.
- **Tension on the catheter adds nothing** (Fruhman, 140 women).
- **Parkland (Adhikari)**: adding **Foley** to oral misoprostol → **no gain in vaginal delivery (77% with misoprostol alone), but 30% more chorioamnionitis**.

Hygroscopic cervical dilators

- **Laminaria japonicum**, synthetic (**Dilapan-S**) (Ch 11).
- Ascending infection **not verified** → appear safe.
- Need a **speculum and an exam table**. **Few benefits over prostaglandins**.

5. Methods of Induction and Augmentation

- Oxytocin used for **>70 years** (Theobald 1948). Others: prostaglandins (misoprostol, dinoprostone) and mechanical methods (stripping, amniotomy, EASI, balloons, dilators).
- Each unit should have **written protocols** (Guidelines for Perinatal Care).
- ⚠️ **Prostaglandins for augmentation are considered experimental** (high tachysystole).

Misoprostol for induction

- ROM at or near term, or favorable cervix: **100 µg oral or 25 µg vaginal ≈ IV oxytocin; oral may be superior**.
- More tachysystole (especially higher doses); may still need oxytocin. Either drug is suitable.
- **Parkland protocol**:
 1. **100 µg oral**
 2. Repeat after **6 h** if labor inadequate
 3. **Oxytocin** (if needed, for hypotonic labor) **4 h after the 2nd dose**, or if tachysystole occurs

- **Misoprostol vaginal insert (up to 24 h):** shorter time to delivery but **more cesareans and adverse neonatal outcomes** vs oral; mitigated if limited to **10 h**.
- **Augmentation (Bleich RCT):** oral **75 µg every 4 h, max 2 doses** = safe and effective. More tachysystole, but **no difference in nonreassuring status or cesarean** vs oxytocin.

6. Oxytocin

- Ripening and induction are a **continuum**. Oxytocin augmentation is key to **active management of labor** (Ch 22).
- ACOG: **monitor FHR and contractions** (palpation or electronic).

IV administration

- **1-mL vial = 10 units.** Add **10 or 20 units (10,000 or 20,000 mU)** to **1000 mL** → **10 or 20 mU/mL**, by **infusion pump**.
- Piggyback into the main line **close to the venipuncture site** (avoid boluses).
- Goal: contractions enough for cervical change and descent **without a nonreassuring (category III) tracing**.
- **Stop oxytocin if: >5 contractions in 10 min, >7 in 15 min**, or a **persistent nonreassuring FHR**.
- **Half-life 3–5 min** → stopping lowers contraction frequency rapidly.
- **Uterus responds in 3–5 min; plasma steady state in 40 min.**
- Response varies (uterine activity, cervix, GA, individual). **Uterine sensitivity rises from 20 to 30 wk and rapidly at term.**
- **Cochrane:** failure to deliver vaginally within 24 h **8% with oxytocin vs 54% expectant**.

Table 26-5. Various Low- and High-Dose Oxytocin Regimens Used for Labor Induction

Regimen	Starting Dose (mU/min)	Interval (min)	Incremental Increase (mU/min)
Low-dose	0.5–1.5	15–40	1–2
	2	15	4, 8, 12, 16, 20, 25, 30
High-dose	4	15	4
	4.5	15–30	4.5
	6	20–40 ^a	3–6 ^b

^aUterine tachysystole is more common with shorter intervals. ^bWith uterine tachysystole and after oxytocin infusion is discontinued, it is restarted at one half the previous dose and then increased at 3-mU/min incremental doses. From ACOG, 2019a; Hauth, 1986; Satin, 1992, 1994; Son, 2021; Tesemma, 2020.

Regimens — the evidence

- Originally only low-dose in the US. **Dublin (O'Driscoll):** start **6 mU/min**, ↑ **by 6 mU/min**. 1990s trials: **high-dose (4–6) vs low-dose (0.5–1.5 mU/min)**.
- **Satin (Parkland), 6 vs 1 mU/min:**
 - **Induction (1112 women):** shorter admission-to-delivery, **fewer failed inductions, no neonatal sepsis**
 - **Augmentation (1676 women):** faster delivery, **fewer forceps, fewer cesareans for dystocia, less chorioamnionitis and neonatal sepsis**
 - Tachysystole → stop, **restart at half the stopping dose**, then ↑ **by 3 mU/min**. No adverse neonatal effects.
- **Son (2021) RCT confirmed** these findings. Parkland has used **6 mU/min start and increments since 1990**; other units use **2 mU/min**. Either regimen serves for induction or augmentation.

Dose interval

- Ranges **15–40 min**. **Satin (6 mU/min regimen): 20-min vs 40-min increases** → **cesarean for dystocia 8% vs 12%**, but **more tachysystole with 20 min**.

- Others: **4 mU/min every 15 min** (Frigoletto, Xenakis); **4.5 mU/min every 30 min** (Merrill and Zlatnik); **6 mU/min every 15 min** (López-Zeno).

Maximum dose

- Varies by woman. **Wen (1151 nulliparas)**: likelihood of vaginal delivery ↓ at and beyond **36 mU/min**, but at **72 mU/min half still delivered vaginally**.
- If contractions are **<200 MVU**, fetal status reassuring and labor arrested → doses **>48 mU/min have no apparent risk**.

Risks vs benefits

Uterine rupture

- **Rare without a scar**, even in parous women:
 - Flannelly: **0 ruptures in 27,829 nulliparas**; **8 in 48,718 parous women**, only **1** with oxytocin
 - Denmark: **3.3 per 100,000** women without prior cesarean (highest in multiparas)
- **Parkland (8 yr, ~95,000 births)**: **15 primary ruptures, 14 with oxytocin**; half had prostaglandins before oxytocin.

Water intoxication

- Oxytocin is homologous to **arginine vasopressin** → **antidiuretic**.
- At **≥20 mU/min, free water clearance falls markedly**. With large aqueous fluid volumes → **water intoxication** → **convulsions, coma, death**.
- **Prevention**: for prolonged high doses, **increase the concentration rather than the flow rate**; use **normal saline or lactated Ringer's**.

Uterine contraction pressures

- Spontaneous labor: **90–390 MVU** (Ch 24). A mean/median spontaneous pattern of **140–150 MVU** led to vaginal delivery.
- **Hauth protocol (active-phase arrest)**: **>90% reached ≥200–225 MVU**. Nearly all with persistent arrest despite oxytocin generated **>200 MVU**. **No adverse maternal or perinatal effects** in those who needed cesarean.
- **No safety data** for contraction patterns with **prior cesarean, twins or an overdistended uterus**.

7. Amniotomy for Induction and Augmentation

For induction ("surgical induction")

- **Always implies a commitment to delivery**.
- Main drawback: **unpredictable, sometimes long interval to labor onset**.
- Bakos: amniotomy alone or with oxytocin was **superior to oxytocin alone**.
- **Mercer (209 induced women)**: **early amniotomy (1–2 cm) vs late (5 cm)** → **labor 4 h shorter**, but **more chorioamnionitis**.
- Meta-analysis (De Vivo 2020): early amniotomy → **shorter induction-to-delivery, no ↑ cesarean**.

For augmentation

- Common when labor is abnormally slow. **Amniotomy at ~5 cm shortens spontaneous labor by 1–1½ h**, without ↑ oxytocin need or overall cesarean rate, and **no adverse perinatal effects**.

Table 26-6. Randomized Clinical Trials of Elective Amniotomy in Early Spontaneous Labor at Term

Study	Number	Mean Dilation at Amniotomy	Mean Shortening of Labor	Need for Oxytocin	Cesarean Delivery Rate	Abnormal Tracing	Neonatal Effects
Fraser (1993)	925	<5 cm	125 min	None	None ^a	None	None
Garite (1993)	459	5.5 cm	81 min	Decreased	None	Increased ^b	None
UK Amniotomy Group (1994)	1463	5.1 cm	60 min	None	None	NA	None

^aNo effect on overall rate; cesarean delivery for fetal distress significantly increased. ^bIncreased mild and moderate umbilical cord compression patterns. NA = not assessed. Note: this footnote conflicts with the text, which says cesarean rates for fetal distress were not greater after amniotomy.

- **Rouse (1994): amniotomy + oxytocin for active-phase arrest → 44 min faster** than oxytocin alone; **delivery route unchanged**; ↑ **chorioamnionitis**.

Technique and safety

- ⚠️ **Cord prolapse risk** → **don't disengage the head**; fundal and/or suprapubic pressure helps. Some rupture **during a contraction**.
- Head poorly applied → controlled release: several punctures with a **26-gauge needle** in a **ring forceps** under speculum view, or guided through the **pudendal-block trumpet**. (The membranes often tear anyway.)
- **Check FHR before and immediately after amniotomy** (cord prolapse, rarely **abruption**).

8. Membrane Stripping (Sweeping)

- Common. Several studies: **safe**, ↓ **postterm pregnancy**, without consistently raising ROM, infection or bleeding.
- But RCTs are limited; a **meta-analysis of two found sweeping ineffective**.
- Downsides: **discomfort and bleeding**.

Quick-Recall: Ripening and Induction Agents

Agent	Dose	Key rules	Main risk / note
Dinoprostone gel (Prepidil)	0.5 mg intracervical	q6h, max 3/24 h ; recline 30 min; oxytocin 6–12 h later	Tachysystole 1–5%
Dinoprostone insert (Cervidil)	10 mg posterior fornix, 0.3 mg/h	Recline 2 h; remove at 12 h /labor; oxytocin ≥30 min after removal	Removable if tachysystole
Dinoprostone 20-mg suppository	—	Not for ripening ; termination 12–20 wk, IUFD ≤28 wk	—
Misoprostol vaginal	25 µg q3–6h	ACOG dose; off-label	Tachysystole >25 µg; meconium
Misoprostol oral	50–100 µg q3–6h (Parkland 100 µg, repeat at 6 h, oxytocin 4 h later)	Augmentation: 75 µg q4h × 2	Higher Apgar, less PPH than vaginal
Prostaglandins in general	—	Not with prior uterine incision ; not in spontaneous labor	Uterine rupture
Foley ± EASI	30-mL balloon; saline 30–40 mL/h	Unfavorable cervix; intact or ruptured membranes	Less tachysystole; EASI ↓ chorioamnionitis (6% vs 16%)
Laminaria / Dilapan-S	Intracervical	Speculum + table	Few benefits
Nitric oxide donors	—	Less effective	Headache, nausea, vomiting
Oxytocin low-dose	0.5–2 mU/min	↑ 1–2 mU/min q15–40 min	—
Oxytocin high-dose	4–6 mU/min	↑ 3–6 mU/min q15–40 min; tachysystole → restart at ½ dose, ↑ 3	Water intoxication ≥20 mU/min
Amniotomy	—	Commits to delivery; check FHR before/after	Cord prolapse, chorioamnionitis

Key Numbers to Memorize

- Induction rate **9.5% → 27%** (1991 → 2019)
- ARRIVE: **>6000** nulliparas at **39 wk**; cesarean **18.6% vs 22.2%**; hypertensive disorders **9.1% vs 14.1%**
- Induction linked to **17%** of unanticipated peripartum hysterectomies
- Elective induction vaginal delivery: **97%** multiparas, **76%** nulliparas
- Favorable: BMI **<30**, birthweight **<3500 g**
- Failed induction: allow latent phase **18 h**; **≥12 h** oxytocin after ROM; **15 h** multiparas
- Bishop **>8** favorable; **≤6** unfavorable
- Prepidil **0.5 mg** q6h ×3; Cervidil **10 mg**, **0.3 mg/h**, out at **12 h**; oxytocin **6–12 h** after gel, **30 min** after insert
- PGE₂ tachysystole **1–5%**; Cochrane: tachysystole with FHR changes ×3
- **Tachysystole = >5 contractions/10 min**; stop oxytocin at **>5/10 min** or **>7/15 min**
- Misoprostol **25 µg** vaginal; **100 µg** oral (Parkland); **75 µg** oral augmentation
- EASI chorioamnionitis **6% vs 16%**; Foley + misoprostol → **30%** more chorioamnionitis, no VD gain (**77–78%**)
- Oxytocin: **10 U/mL** vial; **10–20 mU/mL** infusate; **t½ 3–5 min**; response **3–5 min**; steady state **40 min**
- Oxytocin vs expectant: undelivered at 24 h **8% vs 54%**
- 20- vs 40-min increments: dystocia cesarean **8% vs 12%**
- Max dose: VD falls **≥36 mU/min**; **half** deliver at **72 mU/min**; **>48 mU/min** OK if <200 MVU and reassuring
- Water intoxication at **≥20 mU/min**

- Rupture: **0/27,829** nulliparas; Denmark **3.3/100,000**; Parkland **14 of 15** with oxytocin
- Spontaneous labor **90–390 MVU**; target **200–225 MVU**
- Amniotomy: early (1–2 cm) → **4 h** shorter; at ~5 cm → **1–1½ h** shorter; with oxytocin for arrest → **44 min** shorter